RythuSaathi Complete Implementation Plan

A gated delivery roadmap for the frozen six-agent, all-India advisory architecture

Planning basis This plan translates the approved architecture into sequenced implementation work, acceptance gates, and release criteria. It does not narrow product scope to a demonstration crop or region. Coverage is added from verified data and evidence, and every capability reports its actual support status.

Architecture is frozen: Microsoft Agent Framework is mandatory; Groq is the primary LLM provider; exactly six agents remain; multi-user remembered BYOK uses KMS-backed envelope encryption.

1. Delivery objectives and boundaries

Deliver a secure, evidence-grounded agricultural advisory service for users selecting a state, district, and crop dynamically. Implement the approved architecture in increments: first prove the framework/provider route, then secure identity and credentials, establish registries and evidence contracts, integrate data sources, implement each agent, and release only after reliability and security gates pass.

This roadmap is a dependency-driven sequence rather than a fixed calendar promise. Staffing, source access, KMS availability, and the framework spike determine elapsed time. Coverage should grow through configuration and curated data updates, not workflow branches for named locations.

- Keep Intake, Weather, Crop, Market, Scheme, and Coordinator as six separately identifiable agents coordinated by Microsoft Agent Framework graph workflows.

- Keep business logic, deterministic tools, source adapters, provider clients, and user interfaces independently testable.

- Never fabricate missing data or silently substitute a crop, district, commodity, or credential source.

- No implementation begins until the framework compatibility spike and security design gates in this plan are accepted.

2. Delivery sequence and decision gates

A phase may begin when its entry conditions are met. A failed gate returns the work to its owning phase with a documented blocker; it does not authorize an unapproved framework or scope change.

Phase	Primary outcome	Exit gate
0. Readiness and feasibility	Verified Python/Microsoft Agent Framework/Groq route and finalized implementation choices.	All spike criteria pass; exact package/platform set recorded; no unresolved concrete blocker.
1. Security and platform foundation	Authenticated tenancy, credential vault design, service boundaries, environments, telemetry controls.	Threat model and credential lifecycle approved; isolation and non-disclosure tests pass.
2. Context and evidence contracts	Versioned crop/geography/commodity registry contracts and provenance model.	Context resolution, ambiguity, unsupported status, and provenance contract tests pass.
3. Data and knowledge adapters	Weather, market, crop knowledge, and central/state scheme retrieval adapters.	Each adapter has live/cached/historical/fixture status, timestamps, failure behavior, and source citations.
4. Six-agent workflow	Six agents operating on one board in the required sequential graph.	Context and write-scope tests pass; all findings are sourced or explicitly unavailable.
5. Product interfaces	Authenticated interface(s) to select context, manage BYOK key, and view the advisory.	No key enters client persistence/telemetry; end-to-end flows pass for complete and partial data.
6. Hardening and release	Security, reliability, operations, and staged coverage release.	Go-live checklist passes; rollback and key-revocation procedures are rehearsed.

3. Phase 0 - Readiness and feasibility spike

Purpose: verify the required orchestration stack and Groq integration before package pinning or broad implementation. Keep the spike isolated and disposable. It validates APIs and compatibility; it is not an application scaffold or permission to build product features.

Work items

- Choose a supported Python 3.11+ runtime candidate and intended operating platform. Inspect the current Microsoft Agent Framework package metadata, release notes, and the exact core/OpenAI integration dependencies.

- Build a minimal graph workflow with six named agent identities in Intake -> Weather -> Crop -> Market -> Scheme -> Coordinator order. Pass one canonical context object through each step and append typed findings to a board-like test object.

- Configure the Microsoft Agent Framework Python OpenAI-compatible client to reach Groq through its documented endpoint; test configurable model selection and one provider tool/function call.

- Exercise success, timeout, rate-limit, malformed response, and authentication failure paths. Capture supported graph/workflow events and normalize failures into planned statuses.

- Inject a test user credential only through the candidate backend credential boundary. Use a secret canary to inspect prompts, agent inputs, outputs, logs, traces, event payloads, and board state.

Acceptance criteria

1. Candidate packages install and import together on the chosen Python/platform combination without unresolved dependency conflicts; exact versions and hashes/lock strategy are recorded for later approval.

2. The current supported graph workflow executes six distinct agent identities in order and passes the same selected context without leakage or mutation by unauthorized steps.

3. Groq inference and required tool/function behavior work with a configurable model identifier; provider failures and workflow events are observable and mapped to the planned error model.

4. The credential canary is absent from all agent inputs/prompts, logs, traces, queues, caches, analytics, board records, errors, and responses; only the protected provider credential path accesses it.

5. No selected API is experimental or preview for a required capability unless explicitly accepted as a documented risk. Any blocker has a minimal reproduction and a proposed resolution.

If the spike fails, diagnose against Microsoft Agent Framework documentation and supported APIs. Do not swap frameworks. Present the concrete blocker and evidence for approval before proposing any framework change.

4. Phase 1 - Security and platform foundation

Purpose: establish trust boundaries before storing a user secret or accepting production user data. Confirm deployment identity, user authentication, tenant claims, KMS, credential store, retention, backup, and operational ownership.

Implementation work

- Create an authenticated backend boundary. Derive tenant/user identity only from verified authentication claims; never trust tenant/user IDs supplied by a client request body.

- Define a credential service that supports create, status, use, rotate, revoke, and delete. Persist authenticated ciphertext and wrapped DEK with tenant/user binding and minimal non-secret metadata; keep it separate from the AdvisoryBoard and ordinary application records.

- Use envelope encryption: authenticated encryption with a data key, with that data key wrapped by a KMS-managed key. Restrict unwrap/encrypt permissions to the backend service identity; separate KMS administration from routine service operations.

- At inference, authorize the caller, retrieve only that caller's active record, unwrap and decrypt just in time within the backend provider/credential layer, pass the transient key to the provider client, then discard it as soon as practical.

- Return only a non-secret credential status to the frontend. Redact credential-bearing request fields before logs, traces, metrics, exception reporting, analytics, and support tooling.

- Implement immediate revocation, rotation to a replacement credential, deletion, KMS key rotation/rewrapping, and documented backup-deletion behavior. Audit actions without recording key values, ciphertext, or reconstructive key-derived values.

- Enforce explicit provider credential mode. A request selecting BYOK cannot fall back to deployment credentials after missing key, revocation, authentication failure, quota exhaustion, or provider error.

Security acceptance gate

Pass cross-tenant authorization tests, secret canary non-disclosure checks, KMS least-privilege review, rotation/revocation/deletion tests, and recovery/backup review. Security review must approve the credential threat model and operational runbook before user BYOK is enabled.

5. Phase 2 - Context, registry, and evidence contracts

Purpose: make crop/geography breadth data-driven and make source limitations first-class data. Keep product support status separate from biological possibility: a missing mapping means unverified system coverage, not that the crop is absent from a region.

Implementation work

- Define versioned schemas for crop IDs/aliases, state and district IDs/aliases, parent-child relationships, crop-region-season/stage coverage, and crop-to-market commodity mappings.

- Record source, version, validity dates, language, reviewer/maintainer, and update time for registry records. Support additions and corrections without changing agent workflow code.

- Define one validated CaseContext containing canonical state, district, crop, supplied crop stage/season and farmer facts, plus resolution confidence/status and registry versions.

- Define EvidenceRecord fields for source ID/URL, publisher, retrieval time, source publication/observation/issue time, crop/geography scope, units, page/section, source mode, freshness, quality, and typed payload.

- Define statuses for available, unavailable, stale, failed, and unsupported; distinguish source mode (live, cached, historical, fixture) from status and data quality.

- Define shared AdvisoryBoard schemas and append-only write ownership for case, weather, crop guidance, market, schemes, advisory, and trace. Never include credential values, ciphertext, or key-derived material.

Acceptance gate

Registry contract tests cover multiple states, districts, crops, aliases, ambiguous matches, missing mappings, effective-date changes, and invalid references. Unsupported and ambiguous cases must block inappropriate downstream specificity while still allowing clearly scoped general evidence.

6. Phase 3 - Data and knowledge adapters

Purpose: integrate sources behind explicit adapters so source changes do not alter agent contracts. Select source endpoints and access terms during readiness; verify current availability, authentication, response formats, freshness, and usage rights before implementation.

Adapter/workstream	Implementation scope	Minimum acceptance
Weather	Forecasts, rainfall and warning/alert records for canonical geography; preserve provider issue and observation times.	Timeout/empty/malformed responses are explicit. Stale/cache age is shown. No forecast claim without dated source data.
Market	Market/commodity mapping, dated mandi observations, price units and reproducible trend calculation.	Missing market, ambiguous commodity, or no recent observations is reported; no guaranteed best-sale date or invented price.
Crop knowledge	Curated official/authoritative crop guidance retrieved by selected crop and supported geographic scope.	Citation metadata and applicability scope survive retrieval; unrelated region/crop passages are excluded or labeled as general.
Schemes	Central and state official pages/PDFs, page-aware retrieval, scheme version/date and criterion evidence.	Citations point to dated passages; met/unmet/unknown stay criterion-level; never promise eligibility or award.
Cache and source status	Adapter-level cache metadata, data mode, freshness and failure state.	Automated tests distinguish live, cached, historical, and fixture. Fixtures can never be labeled live.

Retrieval implementation

Begin with an auditable, page-aware retrieval corpus and lexical retrieval if it meets relevance requirements. Keep ingestion metadata, source revision, superseded documents, and page/section spans. Add more advanced retrieval only when evaluated misses justify it. Do not let the language model infer crop-region applicability that is absent from source metadata.

7. Phase 4 - Agent and workflow implementation

Purpose: implement the frozen six responsibilities as separate framework agents and deterministic tools. The workflow is sequential initially; any later parallelization must preserve board write safety, context consistency, and traceability.

Agent	Build order and key behavior	Acceptance check
Intake	First. Validate required user selection and facts; canonicalize state, district, crop; expose ambiguity and coverage.	Ambiguous inputs request clarification; unsupported IDs are explicit; canonical context is immutable to downstream agents.
Weather	Second. Call weather adapter with canonical geography; append observations and provider status.	Correct selected district used; data timestamps and units retained; failure produces no fabricated weather.
Crop	Third. Retrieve evidence for selected crop/region/stage and interpret weather only within evidence scope.	Claims cite evidence IDs; unsupported scope is qualified or withheld.
Market	Fourth. Resolve commodity mapping then retrieve market observations.	Correct mapping and market scope; dated units; trend reproducible; no unsupported timing guarantee.
Scheme	Fifth. Retrieve central and applicable state schemes; compare only facts the farmer supplied.	Passage/page citations and criterion met/unmet/unknown; missing facts remain unknown.
Coordinator	Last. Check required evidence, safety, scope and freshness; compose final response or partial/insufficient result.	Every material claim traceable; unsupported claims suppressed; stale/unavailable evidence disclosed; safety notices prioritized.

Every tool call has a meaningful operational purpose, typed inputs/outputs, timeouts, and explicit error states. Framework events and application trace records identify agent/tool, case ID, timestamps, duration, status, and source/evidence IDs without carrying secrets.

8. Phase 5 - User interfaces and user journey

Purpose: expose the same backend workflow through the approved interface scope once confirmed. Interface choice does not change context or agent contracts.

1. Authenticate user and establish tenant context before showing saved credential status or user-specific records.

2. Collect or confirm state, district, and crop from dynamic registries. Surface search aliases, ambiguity, and unsupported coverage without silently selecting a nearby option.

3. Collect only farmer facts needed for the current advice or scheme criteria; show why they are requested and follow the approved consent/retention policy.

4. Provide a secure BYOK key-entry and update flow. Never prefill a saved key, return it, or persist it in browser storage. Show a non-secret configured/revoked/error status.

5. Run the workflow and present each category with source mode, freshness, citations, geographic/crop scope, and limitations. Clearly label fixture data in non-production environments.

6. Allow correction of crop/location and rerun the case using a new context/board. Keep case isolation and avoid carrying findings into a different crop-region context.

9. Phase 6 - Integration, quality, and security verification

Purpose: prove the system works across combinations and degraded states, not just a happy-path sample. Automated suites use deterministic fixtures and do not make live network calls.

Test suite	Coverage	Release threshold
Context matrix	Multiple crops across states/districts; same crop in multiple places; multiple crops in one place; no cross-case leakage.	All six agents see exact canonical context and expected evidence filters.
Resolution and coverage	Ambiguous aliases, unsupported crop-region combinations, absent registry entries, stale mappings.	Clarification/unsupported result; no guessed substitutions or false local specificity.
Data reliability	Live error, timeout, malformed response, stale cache, historical evidence, conflict, no result, fixture-only.	Correct mode/status and dates; dependent claims withheld or qualified.
Scheme correctness	Central/state scheme retrieval, superseded document, each criterion met/unmet/unknown.	Dated citations and unknowns retained; no eligibility guarantee.
Credential security	Tenant isolation, owner checks, rotation, revoke, delete, KMS errors, invalid key and no fallback.	Unauthorized calls denied; BYOK failure never uses deployment key; secret canary absent from every forbidden surface.
Workflow integrity	Six-agent ordering, authorized board writes, partial result, agent/tool failure, retry/timeout handling.	Trace complete; findings append-only; Coordinator returns safe partial or insufficient result.
Advisory quality	Claim-to-evidence audit, applicability, freshness, uncertainty, safety and readability.	No unsupported material claim; citations and limitations rendered for all relevant recommendations.

10. Phase 7 - Deployment readiness and staged coverage

Purpose: release the system in a controlled way while keeping national product scope separate from the actual coverage available at launch. No one crop-region pair becomes an architectural restriction; launch support is a registry and source-readiness fact.

- Complete production threat model, security review, privacy/data retention review, and provider/source access approvals.

- Configure deployment secrets using approved environment/secret-manager paths; validate no secrets in images, source, logs, traces, or build artifacts.

- Publish operational dashboards for provider status, source freshness, workflow failures, unsupported coverage, and credential lifecycle events without personal secrets.

- Document key rotation, user key revocation, incident response, data deletion, provider outage, stale-data, and rollback procedures; rehearse them.

- Enable only verified source/registry coverage. Make unsupported combinations explicit and support registry updates without code deployment where safe.

- Use feature flags and staged traffic rollout; monitor citation failures, stale-source rates, context-resolution errors, provider error rates, and safety escalations.

11. Cross-cutting workstreams and ownership

Workstream	Accountable owner	Key artifacts
Architecture and workflow	Technical lead	Framework spike report; graph/workflow contracts; six agent definitions; ADRs.
Identity and BYOK security	Security/backend lead	Threat model; tenant authorization; KMS design; credential service; lifecycle runbook.
Agricultural evidence and registries	Data/agronomy lead	Crop/geography/commodity registries; source inventory; regional applicability rubric; review cadence.
Provider and data adapters	Backend/data integration lead	Groq adapter; weather/market/scheme connectors; retries/timeouts; provenance and freshness behavior.
Testing and quality	QA/engineering lead	Fixture matrix; security canary suite; claim-evidence checks; regression and release reports.
Product and operations	Product/operations lead	Interface scope; consent/retention decisions; rollout plan; support and incident procedures.

For a small team, one person may own several workstreams, but security review, credential access, and source correctness still need explicit review ownership.

12. Operational decisions required before affected work

Decision	Needed before	Decision owner/output
Microsoft Agent Framework candidate versions, Python/platform, spike approval	Broad implementation	Technical lead: passing spike record and pinned candidate dependency set.
Authentication and tenant identity claims	Credential persistence	Identity/backend owner: verified subject/tenant mapping and authorization rules.
KMS, credential store, backup/deletion, key rotation	BYOK storage	Security/platform owner: approved services, roles, encryption metadata, lifecycle policy.
Initial Groq model ID, limits, timeouts and retry policy	Provider production configuration	Provider owner: deployment/user config model and reviewed service limits.
Source endpoints, access terms, owners and update cadence	Each live adapter	Data owner: source register and operational readiness evidence.
Initial registry corpus, languages, regional applicability rubric	Coverage claims	Agronomy/data owner: reviewed versioned records and support labels.
Freshness limits, cache behavior, conflict handling	Any user-facing live/cached data	Product/data owner: category-specific display and suppress/qualify rules.
Farmer data collection, consent, retention, interface/language scope	Interface and scheme matching	Product/privacy owner: approved fields, purposes, locales and retention.

13. Key risks and mitigations

Risk	Impact	Mitigation / gate
Framework or Groq compatibility mismatch	Schedule delay or unsupported tools/events.	Complete the bounded spike before package pinning; use current supported graph API; document reproducible blockers.
Credential exposure or cross-tenant access	Compromise of user provider accounts.	KMS envelope encryption, strict tenant binding, least privilege, no-fallback rule, canary tests and security review.
Sparse or uneven regional evidence	Advice appears more localized or certain than sources justify.	Applicability metadata, registry coverage status, source citations, explicit unsupported output, staged evidence onboarding.
Provider/API availability or limits	Partial responses and request failures.	Adapters with timeout/rate-limit handling; explicit data status; suppress dependent claims; monitor provider errors.
Stale or conflicting data	Unsafe or misleading current recommendations.	Category-specific freshness rules; source timestamps; conflict state; qualify or withhold claims.
Scheme changes or missing farmer facts	Incorrect eligibility impression.	Dated official material; criterion-level unknown; no guarantee; prompt only for approved necessary facts.
Scope expansion without source readiness	Unverifiable claims across nominal all-India scope.	Keep product model all-India but enable only verified coverage; coverage state is dynamic and visible.

14. Definition of done

- The required framework spike passed and its exact package/platform choices are recorded; Microsoft Agent Framework remains in use.

- All six agents execute in the required workflow and preserve selected context, authorized board writes, and source traceability.

- User BYOK records are tenant/user-bound and KMS envelope-encrypted; use is just in time and protected; rotation, revocation, deletion, no-fallback, and non-disclosure checks pass.

- Registry-driven tests prove multiple crop-region combinations, ambiguous selections, unsupported cases, and independent coverage updates.

- Live/cached/historical/fixture data are distinguishable, and failure/staleness/conflict tests prove unsupported claims are withheld or qualified.

- Central and state scheme results are cited and criterion-level; weather, market, crop guidance, and scheme outputs retain provenance and applicable scope.

- Operational runbooks, monitoring, rollout/rollback procedures, privacy/retention policies, and accountable owners are approved.

- No unresolved high-severity security, safety, data integrity, or framework compatibility issue remains open.

15. Architecture references

Microsoft Agent Framework Python compatibility and releases: https://github.com/microsoft/agent-framework/releases

Microsoft Agent Framework workflow concepts: https://learn.microsoft.com/en-us/agent-framework/concepts/workflows/

Microsoft Agent Framework OpenAI-compatible endpoint configuration: https://learn.microsoft.com/en-us/agent-framework/hosting/self-hosting/openai-endpoints

Groq OpenAI compatibility: https://console.groq.com/docs/openai

The compatibility spike must re-check these sources and the selected package documentation when implementation is authorized; current API behavior can change.